

Training GPT-2 on SEC-EDGAR Financial Filings: A 124M Parameter Language Model for Corporate Disclosure Generation

Li Zhiwei

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Abstract

We present a 124-million parameter GPT-2 language model trained on 1.55 billion tokens of SEC-EDGAR financial filings using the nanoGPT framework. The model was trained on a single NVIDIA RTX 4070 GPU over approximately 5 days, reaching a validation loss of 2.45 at step 37,000 of 47,400 total steps (78% completion). We evaluate the model’s generation quality across multiple SEC filing sections including business descriptions, management discussion and analysis, risk factors, financial notes, and proxy statements. Our analysis reveals that the model successfully learns SEC document structure, financial vocabulary, and boilerplate language patterns, but exhibits characteristic limitations in long-range coherence, numerical consistency, and table extension. We discuss the implications for using small language models in financial document generation and the trade-offs between model size, training data quality, and generation fidelity.

1 Introduction

The proliferation of large language models (LLMs) has created significant interest in domain-specific text generation. Financial filings submitted to the U.S. Securities and Exchange Commission (SEC) through the EDGAR system represent a particularly rich corpus of structured, domain-specific text. These documents follow strict formatting conventions and employ specialized financial vocabulary, making them an

interesting testbed for evaluating language model capabilities in structured document generation.

While state-of-the-art models like GPT-4 and Claude demonstrate impressive general-purpose capabilities, the cost and computational requirements of training and deploying such models limit their accessibility. In this work, we investigate whether a smaller, more accessible model—a 124M parameter GPT-2—can learn to generate convincing SEC filing text when trained on a focused corpus of EDGAR documents.

Our contributions include:

- Training a 124M parameter GPT-2 model on 1.55 billion tokens of SEC-EDGAR filings using consumer hardware
- Systematic evaluation of generation quality across five distinct SEC filing sections
- Analysis of long-range coherence, numerical consistency, and structural fidelity
- Discussion of practical applications and limitations for financial document generation

2 Related Work

2.1 Language Models for Financial Text

Recent work has explored applying language models to financial domains. BloombergGPT [1] trained a 50B parameter model on a mixed corpus including financial data. FinGPT [2] proposed open-source financial language models. However, these efforts focused on much larger models and broader financial tasks.

2.2 Small Language Models

The original GPT-2 [3] demonstrated that language models could generate coherent text at various scales (124M to 1.5B parameters). More recent work has shown that smaller models can achieve surprising performance when trained on high-quality, domain-specific data [4].

2.3 SEC-EDGAR as Training Data

The SEC-EDGAR database contains millions of corporate filings spanning decades. Prior work has used EDGAR data for sentiment analysis [5] and information extraction [6], but little work has explored using EDGAR as a primary training corpus for generative language models.

3 Methodology

3.1 Data Collection and Preparation

We collected SEC-EDGAR filings from the CodeParrot GitHub dataset, which contains processed EDGAR documents. The training corpus consists of:

Metric	Value
Total training tokens	1.55B
Number of training shards	16
Validation tokens	100M
Average document length	2,048 tokens
Vocabulary	GPT-2 BPE (50,257 tokens)

Table 1: SEC-EDGAR training corpus statistics

The data was tokenized using the GPT-2 byte-pair encoding tokenizer and stored as memory-mapped numpy arrays for efficient training.

3.2 Model Architecture

We use the standard GPT-2 architecture with the following configuration:

Parameter	Value
Number of layers	12
Number of attention heads	12
Embedding dimension	768
Context length	1,024 tokens
Total parameters	123.59M
Vocabulary size	50,257

Table 2: GPT-2 124M model configuration

3.3 Training Setup

Training was conducted on a single NVIDIA RTX 4070 GPU with 12GB VRAM using the nanoGPT framework:

Training Parameter	Value
Batch size (micro)	4
Gradient accumulation steps	8
Effective batch size	32,768 tokens
Learning rate	6e-4
Minimum learning rate	6e-5
Warmup iterations	2,000
Maximum iterations	47,400
Weight decay	0.1
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.95$)
Gradient clipping	1.0

Table 3: Training hyperparameters

Training required approximately 5 days to reach step 37,000 (78% completion), achieving a validation loss of 2.45.

4 Results

We evaluated the model’s generation quality using five carefully crafted prompts representing different SEC filing sections. Each prompt contained 500–1000 characters of authentic SEC text, and we generated 1,200 tokens of continuation at temperature 0.7.

4.1 Prompt 1: Business Description (Item 1)

Input: A healthcare SaaS company description with \$487.2M revenue, 8,200 employees, and EHR platform details.

Observations:

- First 2–3 paragraphs maintained healthcare industry context
- Gradually shifted from SaaS to biotech/pharma framing
- After 200 generated tokens, entered a “commercialization of product candidates” loop
- Hallucinated drug names: X-Avent, X-Zentib, S-Zentib, Q-partnerib
- Maintained grammatical structure throughout, even in degenerate sections

4.2 Prompt 2: MD&A (Revenue Analysis)

Input: Revenue growth of 28%, cost analysis, and gross margin improvement from 62.7% to 64.4%.

Observations:

- First continuation paragraph was excellent: “Cost of revenue increased by \$32.1 million, or 26%... primarily attributable to decreased depreciation and amortization expense in the period of the acquisition of DMR”
- Subsequently generated 10 consecutive paragraphs all starting with “Cost of revenue increased/decreased by \$X.X million”
- Numbers became internally inconsistent (e.g., “decreased by \$61.2 million, or 13%, to \$1.9 million from \$2.7 million”)
- Percentage calculations frequently contradicted the dollar amounts cited

4.3 Prompt 3: Risk Factors

Input: Net losses of \$42.3M, \$67.8M, and \$89.1M over three years, with \$523.4M accumulated deficit.

Observations:

- Maintained risk factor structure (heading + explanation) for extended generation
- “Product candidates” appeared 47 times in 25 lines
- Recursive self-reference: “Our product candidates may fail to develop, develop and commercialize our product candidates may fail”
- Lost thread of healthcare SaaS, defaulted to generic biotech risk factors

4.4 Prompt 4: Financial Notes (Revenue Recognition)

Input: Revenue disaggregation table with subscription (\$380M), professional services (\$89M), and hardware (\$18M) revenue.

Observations:

- Table echoed with perfect formatting and alignment
- All numbers reproduced exactly as in the input
- Attempted to start a second table but generated only blank spaces
- Only 50 tokens of meaningful continuation before collapse

4.5 Prompt 5: Proxy Statement (Executive Compensation)

Input: Executive compensation table with three named officers and full compensation breakdown.

Observations:

- Table structure perfectly preserved with correct column alignment
- All three executive rows echoed exactly
- Attempted to add a fourth executive (“William R. Gras”) but only generated partial row

- Immediately collapsed to blank spaces after incomplete row

5 Analysis

5.1 Echo vs. Generate Distinction

A clear pattern emerges: the model excels at **echoing** input content (tables, numbers, formatting) but struggles to **generate** new content that maintains consistency. This suggests the model learned surface-level patterns rather than underlying data relationships.

5.2 Loop Attractors

Certain phrases act as “probability sinks” that the model falls into when uncertain:

- “Commercialization of product candidates” (Prompts 1, 3)
- “Cost of revenue increased by \$X million” (Prompt 2)
- “Raise additional capital” (observed in earlier tests)

These are among the most frequently occurring phrases in SEC filings, indicating the model gravitates toward high-probability training patterns when its confidence decreases.

5.3 Numerical Coherence

- **Dollar amounts:** Plausible scale (\$1M–\$500M range) but internally inconsistent
- **Percentages:** Often don’t match the dollar changes cited
- **Dates:** Consistent (always “December 31, 2023/2022”)
- **Implication:** The model learned the *format* of numbers, not their *meaning*

5.4 Domain Drift

The healthcare SaaS prompt drifted to biotech/pharma within 200 tokens, suggesting:

- Training data may be dominated by biotech 10-K filings

- Biotech risk factors serve as “default” SEC content when the model is uncertain
- Domain-specific fine-tuning data distribution significantly affects generation quality

5.5 Grammar vs. Logic

Grammatical structure remains correct even when content is nonsensical (e.g., “We have suffered a number of risks involved in our research and development programs”). Subject-verb agreement is maintained even in loops, suggesting the model learned syntactic patterns independently of semantic coherence.

6 Discussion

6.1 Practical Applications

The 124M SEC-EDGAR model demonstrates capability for:

- Generating SEC boilerplate language and standard disclosures
- Suggesting section structures and formatting conventions
- Drafting placeholder text that visually resembles authentic filings
- Providing a starting point for human writers to edit and refine

6.2 Limitations

The model is **not** suitable for:

- Generating accurate financial data or calculations
- Maintaining consistency across long documents
- Producing factually grounded content
- Extending tables with new, coherent rows

6.3 Comparison with Larger Models

These limitations are consistent with observations of GPT-2 1.5B (12× larger), which shows similar but less severe patterns. The 124M model’s behavior aligns with ex-

pectations for its parameter count and suggests that:

- Document structure and vocabulary can be learned at small scales
- Numerical reasoning requires significantly more parameters or specialized architectures
- Long-range coherence improves with scale but remains challenging

6.4 Future Work

Potential improvements include:

1. **Data quality:** Curating a balanced corpus across filing types and industries
2. **Model architecture:** Incorporating numerical embeddings or structured prediction heads
3. **Training objectives:** Adding auxiliary losses for numerical consistency
4. **Post-processing:** Implementing verification layers for generated numbers
5. **Scale:** Training larger models (350M, 1.3B) on the same corpus

7 Conclusion

We have demonstrated that a 124M parameter GPT-2 model can learn SEC-EDGAR document structure, financial vocabulary, and boilerplate language when trained on 1.55 billion tokens of corporate filings. The model achieves convincing surface-level generation quality but exhibits fundamental limitations in numerical consistency, long-range coherence, and table extension.

These results suggest that small language models can serve as useful tools for financial document drafting assistance, particularly for generating boilerplate sections and suggesting document structures. However, they cannot replace human expertise for ensuring accuracy and consistency in actual SEC filings.

The SEC-EDGAR corpus presents unique challenges for language models due to its structured nature, numerical density, and strict formatting requirements. Our work provides a baseline for future research on domain-specific financial language models and highlights the trade-offs between model accessibility and generation fidelity.

References

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